

University Depression Rankings Using Reddit Data

NETS 150 - Homework 5 Project

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ABSTRACT

With recent work published on depression analyses of college students within universities across America, debate also emerges around the surrounding phenomenon — which methods are more current and relevant; which metrics are more cohesive; why students experience and express such negative sentiments. This brief study corroborates the results of a study in 2015 conducted on textual sentiment analysis from data gathered from Twitter. The group instead gathers and analyzes data gathered from Reddit subreddits of ivy leagues across the country and Natural Language Processing to support previous results. At the end of the paper, the results show that **while the previous Twitter study finds the University of Pennsylvania to have the highest spot in its depression rankings (followed by Yale, Columbia, etc. amongst the Ivy Leagues), the team's Reddit analysis shows Cornell to lead its own depression rankings (then UPenn, Columbia, Harvard, etc.).** Other results of the study showcase depression similarities between the colleges and words most negatively correlated with depression in each school.

METHODOLOGY

Data set construction

For the construction of the data set to be analyzed, the team scraped all of the main body content of the university reddit posts from the list below and the most recent 10,000 posts of the subreddit r/depression: r/BrownU, r/columbia, r/Cornell, r/dartmouth, r/Harvard, r/princeton, r/UPenn, r/yale, r/depression (for comparison).

The API used for the scraping was the PRAW Reddit Python API available and accessible online.¹ Briefly, the general outline of the functionality of the web scraping code is to create an account to utilize the Reddit API, import PRAW and other relevant libraries, store the university name and the posts of each one in a dataMap (body, time/date, title, comments), then do the same with the most recent 10,000 r/depression posts before converting to a .csv file. The rest of the different functionalities and implementation of the code is shown in the attached file with comments on how to run and use the program.

¹The python reddit api wrapper [1]. (n.d.). Retrieved April 29, 2021, from <https://praw.readthedocs.io/en/latest/>

Sentiment Analysis

After the relevant data needed are scraped from the Ivy League and depression subreddits, the files are passed onto the next file of code running its Natural Language Processing.

The outline of what analysis code does is to visualize the number of posts per subreddit (for scale), time distribution of posts per subreddit (for recency), compute the sentiment values of each post from each subreddit (stores median of reach subreddit's sentiment as well before storing in a dictionary), then visualize the sentiments of each subreddit. Specifically, these include a bar plot of each subreddit sentiment and a violin plot for the distribution of subreddit sentiment.

Lastly, the team included an analysis of the similarity of subreddits to each other through lemmatizing words (grouping similar words together and treating as one i.e. “root” = “roots”), taking cosine similarities, then visualizing through a heatmap and individual subreddit word cloud.

RESULTS

Visualizing the dataset

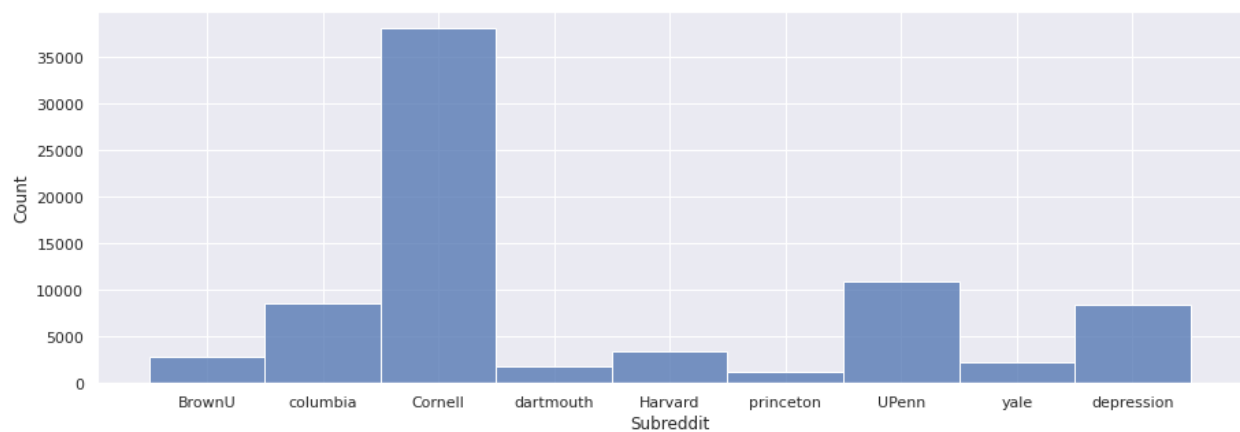


Exhibit 1

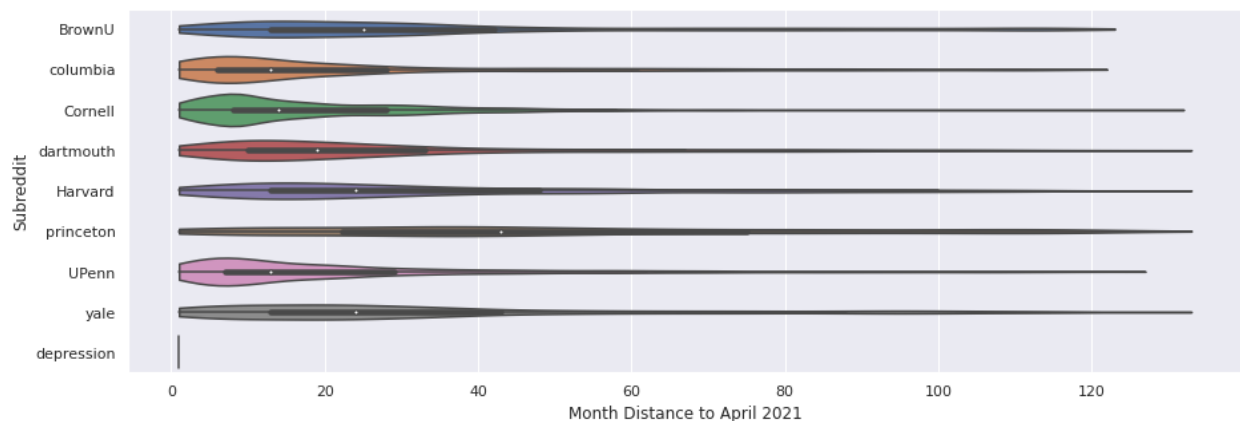


Exhibit 2

Exhibits 1 and 2 visualize the composition of the parts of the data set. Most of the posts come from Cornell, UPenn, Columbia, and r/depression as these subreddits have significantly higher amounts of posts and frequency. It is also important to note the recency of the posts: most are from fewer than 40 months from April 2021. This informs of the timeframe of the student population's sentiment analysis later on.

Visualizing the sentiment

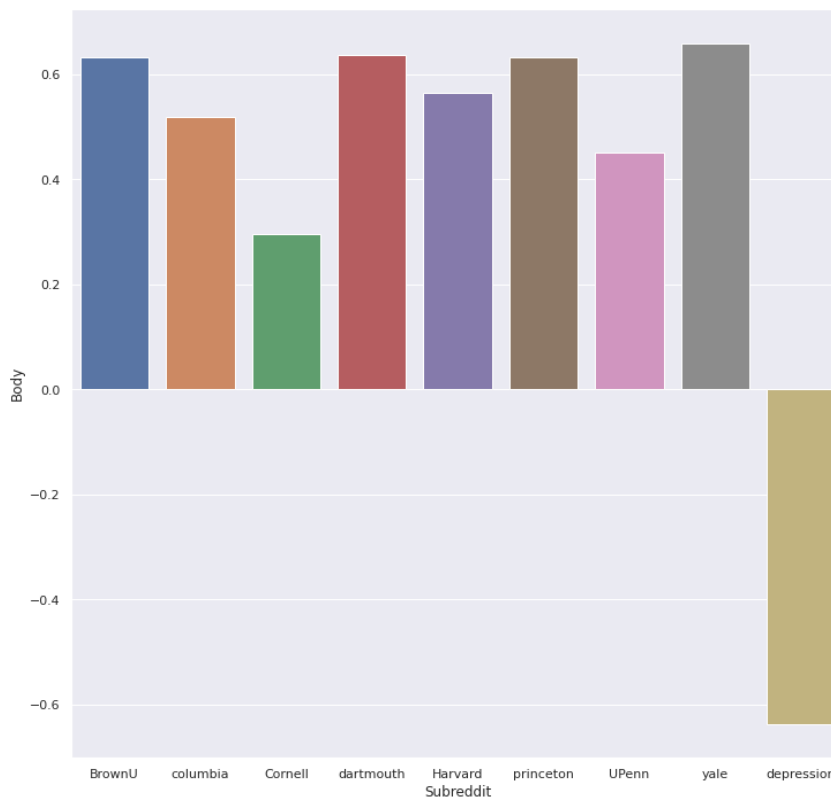
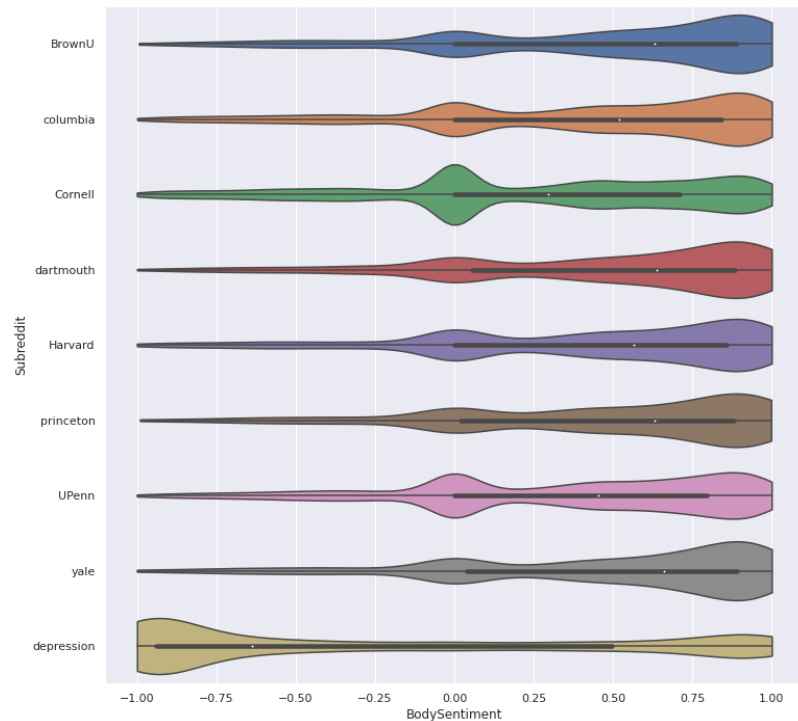


Exhibit 3

Exhibit 3 displays the barplot sentiments of each subreddit from -1 to 1 (most depressed to least). Reasonably so, r/depression has the lowest sentiment score. From the barplot, the lowest sentiment scores come from Cornell, UPenn, Columbia, and Harvard. On the other hand, the highest scores come from Yale, Brown, Dartmouth, and Princeton.

Exhibit 4

The violin plot of Exhibit 4 supports the bar plot but adds a layer of distribution amongst posts. The general trend from this graph is that a bulk of the comments are either neutral or with a high score (happy). This supports Exhibit 3 as the lower end of the scores are “fattest” in the violin plots of Cornell, UPenn, Columbia, and Harvard. The higher end of the scores are consistent with the bar plots as well.



Visualizing the similarity

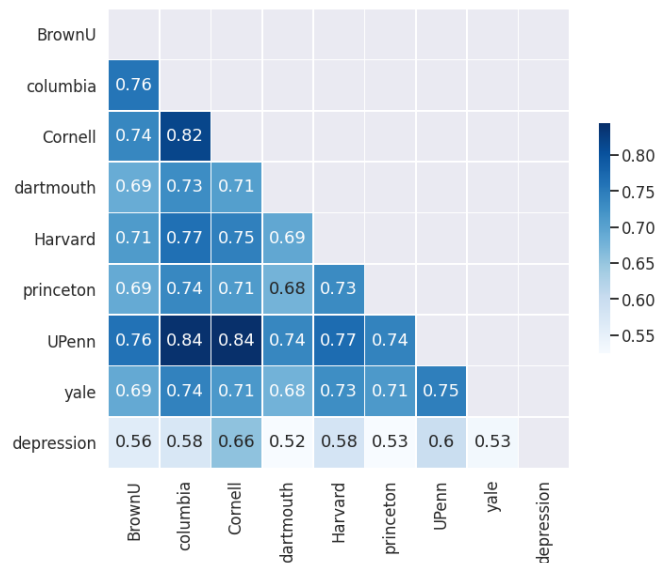


Exhibit 5

Exhibit 5 is a heatmap of cosine similarities between each subreddit. In this exhibit, it is especially important to compare the similarity of each university subreddit with the depression subreddit. In addition to the sentiment analysis of subreddit main body content, the cosine similarities support previous results that indicate the least happy universities are Cornell, UPenn, Columbia, and Harvard.

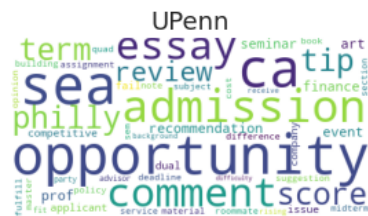
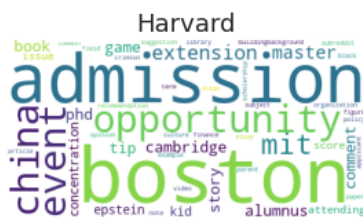
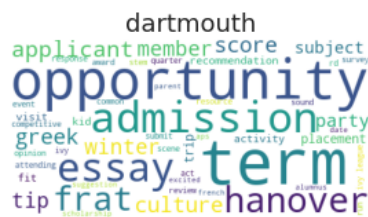
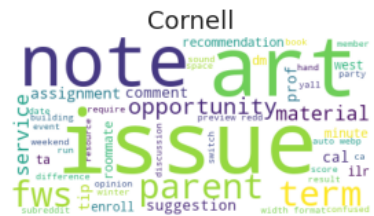
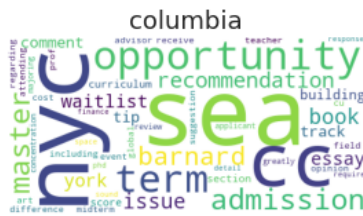


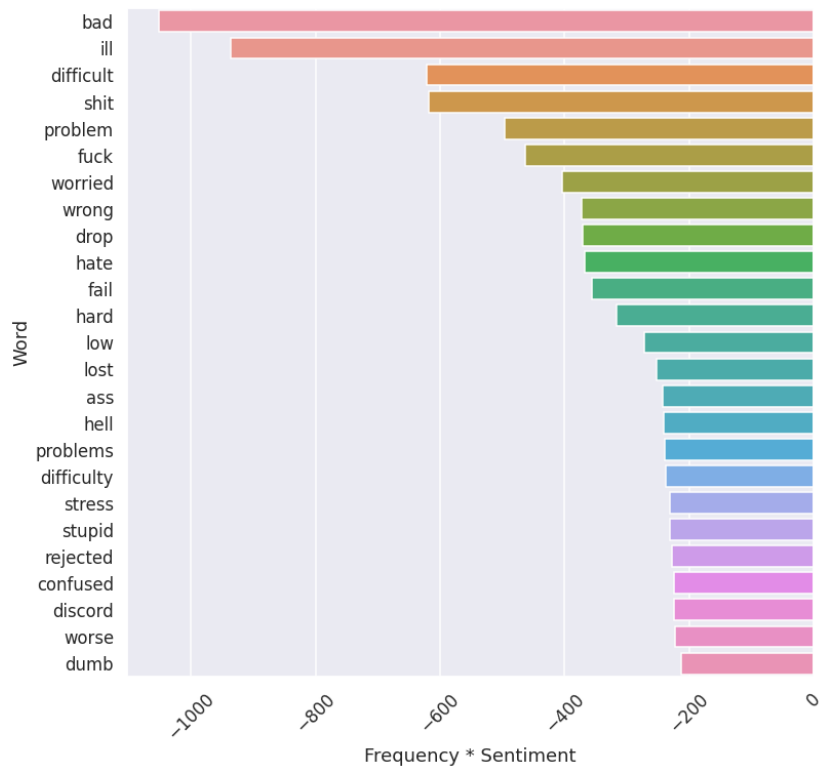
Exhibit 6

Exhibit 6 is a group of word clouds containing the most common words in each subreddit. From these clouds, it is a reasonable inference that the “happier” universities contain an influx of happy posts from celebratory admissions results coming out (from main words including “admission”) which was a significant recent event this April.

From the least happy universities, it is possible that the presence of unfortunate recruiting news influences the median sentiment scores (from word “opportunity”). These are some of the possible conclusions (timely speculations made by team) albeit not necessarily comprehensive nor reflective.

Exhibit 7

Exhibit 7 is a barplot of the most common negative words in the university subreddits. As these words do not conclusively point to any one reason college students are expressing negativity, reasonable speculation can include reasons like a feeling of wanting to drop out (from “drop”), failing at school/ feeling incompetent and lost (“fail”, “stress”, “stupid”, “dumb”, “lost”) or rejection from some form of recruiting (“rejected”).



CONCLUSION

In this study, the team visualizes the sentiments of the university subreddits to have a glimpse of how negative or positive the posts have been, compares these posts to the depression subreddit, and visualizes word clouds to reflect what topics could have possibly led to these sentiments.

The results of the empirical analysis show that among the Ivy League universities, Cornell, UPenn, Columbia, and Harvard seem the most negative in their respective subreddits, while Brown, Dartmouth, Princeton, and Yale seem most positive. These results are inconsistent with the Twitter study that inspired the analysis where UPenn scored the most depressed, followed by Yale, Columbia, then Brown.

This inconsistency however is possibly due to the differences in time frame from which the data was gathered and usage of social media (i.e. it was not considered which platform was more popular with the student populations and therefore more utilized and representative). It is also possible that recent events of either really positive university-wide news or really negative influence the general sentiment of that time (i.e. admissions results being shared in the subreddits in congratulations of new admits).

Sources

Baweja, A., Kong, J., Pan Fang, T., & Wang, Y. (2015). University Depression Rankings Using Twitter Data.

Hill, K. (2019, May 29). Top 50 colleges with the most Depressed student bodies. Retrieved April 30, 2021, from <https://humansofuniversity.com/university-life/top-50-colleges-with-the-most-depressed-student-bodies/>

The python reddit api wrapper¶. (n.d.). Retrieved April 29, 2021, from <https://praw.readthedocs.io/en/latest/>